

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 March 2023

22MCA101 - DATA STRUCTURES WITH ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from each Unit*

		Unit - I			
		Marks	BT*	CO*	PO*
1.	a) Define Abstract Data Type. Implement an ADT for rational numbers.	10	L2	1	1
	b) Write an algorithm to evaluate a postfix expression and apply the same for the postfix expression ABC-D*+E\$F+ and assume A=6, B=3, C=2, D=5, E=1, F=7, where \$ denotes exponentiation Operator.	10	L3	1	2
2.	a) Explain the pointer concept in arrays. Write a C function to find the minimum value stored in an array using pointers.	10	L2	1	1
	b) Write an algorithm to convert an infix expression to postfix expression. Trace with an example.	10	L3	1	2
		Unit - II			
3.	a) Define recursion. Discuss the properties of recursive definition. Explain the overheads involved in recursive functions.	10	L3	2	1
	b) Write a recursive C program to find n th Fibonacci number and trace it for n=3.	10	L2	2	2
4.	a) Write a recursive C code for the following: i) To find the multiplication of two numbers ii) To find the GCD of two numbers. Trace with examples.	10	L3	2	2
	b) Write a C program to implement insert and delete operation of simple queue. Consider queue full and queue empty conditions.	10	L3	2	2
		Unit - III			
5.	a) Define Circular linked list. Implement the queue operations using Circular linked list.	10	L3	3	2
	b) Write a C program to insert a node at the beginning and to delete a node at the end of Singly Linked list.	10	L3	3	2
6.	a) Information of same type is arranged in ascending order in two linked lists. Write a module to combine them into a single, sorted linked list.	10	L3	3	2
	b) Write a C program to insert a node at the beginning and to delete the node at the end of Doubly Linked list.	10	L3	3	2
		Unit - IV			
7.	a) Mention the different methods of Specifying an Algorithm. Explain each.	10	L2	4	1
	b) Discuss about the analysis of algorithm efficiency.	10	L2	4	1
8.	a) Write a recursive procedure to traverse a binary tree using any two traversal methods. Trace it with example.	10	L3	4	2
	b) Briefly explain the various methods of implementing binary trees in C.	10	L2	4	1
		Unit - V			
9.	a) Define Heap. Explain Heap Sort technique with an example.	10	L2	5	1
	b) Compare the following searching techniques: i) Sequential Search, ii) Binary Search, iii) Interpolation Search.	10	L2	5	1
10.	a) Explain the various data structures to represent the graph.	10	L3	5	1
	b) Write Kruskal's minimum spanning tree algorithm. Explain with an example.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
